(MHP Writers)

(ECO 401 Final Term)

(Lesson . 21)

PRICE DISCRIMINATION (PD): when a producer charges different prices for the same product to different customers.

TYPES OF PRICE DISCRIMINATION Three types:

1st degree PD: when identical goods are sold at different prices to each individual consumer. Everyone charged according to what he can pay. **For example,** a bargaining aspect,
2nd degree PD: Different prices charged to customers who purchase different quantities. **Examples,** retail stores,
3rd degree PD: Seller charge different prices to different customers in different markets. **For example,** exporters may charge a higher price in overseas markets if demand is estimated to be more inelastic than it is in home markets. In Pakistan,

PRICE DISCRIMINATION PRE-REQUISITES / CONDITIONS

1. That markets should be independent
2. Firms should have the flexibility to price discriminate
3. Price elasticity of demand for different customers should be different

(MARKET STRUCTURE) (FOUR TYPES):

MONOPOLISTIC COMPETITION: Large number of buyers and sellers and absence of entry barriers. **Example**, Products are differentiated
Long run, a monopolistically competitive firm will make zero economic profit.
Short run, super normal profits are possible
Demand Curve = Downward Sloping

PERFECT COMPETITION: When there are many sellers, there is easy entry and exiting of firms. Sellers are price takers. **Example,** Products are homogeneous or Undifferentiated
Long run, a PERFECT COMPETITION profit less or more and may be firm closed.
Demand Curve = Horizontal Sloping

MONOPOLY: Consists of only one seller or producer. (Barriers to entry)
Long run, a MONOPOLY firm will make Super Normal economic profit.
Demand Curve = Downward Sloping

(Lesson . 22)

OLIGOPOLY: A small number of firms (about 2-20) in the market. A small number of sellers (oligopolists). The word is derived from the Greek for few sellers.

- OLIGOPOLY is aware of the actions of the others.
- It is not possible to identify any single equilibrium in oligopoly.

COLLUSION : Collusion occurs when two or more firms decide to cooperate with each other in the setting of prices and/or quantities.

TWO POSSIBLE SCENARIOS OF OLIGOPOLY :

- **Collusive oligopoly**
- **Non-collusive oligopoly**

COLLUSIVE OLIGOPOLY (CARTEL) WHY Collusion:
(REASONS)

- **Deciding upon market shares**
- **Advertising expenses**
- **Prices to be charged (Identical or Different)**
- **Production Quotas,**

Cartel: A formal (explicit) agreement among firms. Cartels usually occur in an oligopolistic industry, where there are a small number of sellers and usually involve homogeneous products.

FACTORS OF CREATE EASILY COLLUSION:

- Number of firms is small
- Openness production processes
- Product is homogeneous
- Price leader
- Industry is stable
- Government’s strictness in implementing anti-trust
- Against internationally operational cartels
- Collusion is Tacit

NON- COLLUSIVE OLIGOPOLY: If different firms in the oligopolistic structures do not cooperate with each other is known as non collusive oligopoly.

MAXIMIN strategy: Techinque used to maximize the minimum profit usually by some decision

MAXIMAX strategy: Decisions taken under the results in the maximam benifit/Profit

DOMINANT strategy game: One trying to superior on other by their actions/decision

(Lesson . 23)

PRICE STABILITY IN NON-COLLUSIVE OLIGOPOLIES:

KINKED DEMAND CURVE: Explains the "stickiness" of the prices in oligopolistic markets.

(Elastic demand)	(Inelastic demand)
If one firm raises prices, no one else will raise their prices and so the firm will face declining revenues. A relatively more elastic segment for price increases	if one firm lowered its price, everyone else would lower their prices as well and everyone's revenues, including the first firm's revenues would fall A relatively less elastic segment for price decreases

WELFARE ECONOMICS: (Branch of economics dealing with normative issues) The allocation of goods and resources for promoting social welfare. It deals with an economically efficient distribution of resources for the well being of the people.

THE MARGINAL PRIVATE COST OF ADVERTISING: The cost of every additional TV commercial or newspaper advertisement that a firm has to bear.

MARGINAL SOCIAL COST: Social cost (benefit) means the cost (benefit) -- may not be in monetary terms – that is borne by (accrues to) society on the whole.

EXTERNALITY: when the production or consumption of a specific good or service impacts a third party that is not directly related to the production or consumption of that good or service.

A Negative externality occurs when a cost spills over. A positive externality occurs when a benefit spills over.

OPTIMAL LEVEL OF PRODUCTION: Where the marginal revenue (MR) equals the marginal cost (MC).

A tax (subsidy): raises (reduces) prices by shifting the supply curve vertically upwards (downwards).
Market failure: is an imperfection in the price system that prevents an efficient allocation of resources.

MERIT GOOD: Merit good is a good which Govt like people to consume more.
Examples include food stamps, health care, and subsidized housing. If Govt found positive consumption externality then there would be the case of subsidy

PUBLIC GOOD: A commodity or service that is made available to all members of society. Public good is a good that is non-rival and non-excludable.

(Lesson . 24)

Firms are the demanders of the factors and households are the suppliers of the factors. (Mcqs)

The Demand for Factor of Production: (like labor) is a derived demand, because it is "derived" from the goods market.

Leisure: Free time for enjoyable activities and relaxation other than work or necessary sleep.

The Marginal Disutility of Work (MDUW): As the supply of hours of labor increases, wage rate should also be increased.
The marginal disutility of work (MDUW) means the negative impact on the working of laborer for one additional unit of time. The MDUW curve defines the supply curve for labor.

The Opportunity Cost: The opportunity cost of working is leisure (and vice versa) that the worker could have enjoyed during that time had he not been working.

Supply and Demand Curve for Labor: The abour supply curve may bend backwards above a certain wage rate as the income effect of higher wages dominates the substitution effect of higher wages.
The demand curve for labour can be derived from the intersection of the wage rate lines (horizontal parallel lines) and the marginal revenue product of labour (a downward sloping concave function) given by MRPL = MPPL x MRi

The Value of Marginal Product of Labor (VMPL): **(VMPL) = MPPL x Pi It is equal to MRPL when P = MC (as in perfect competition), but otherwise VMPL > MRPL**.

The Net Present Value (NPV): The difference between the present value of cash inflows and the present value of cash outflows over a period of time.
Net present value (NPV) is a financial metric that seeks to capture the total value of an investment Opportunity

Discounting: is the process of converting a stream of future incomes and expenses into a present value. The discount rate is the rate at which the future incomes are discounted. **(FOR Mceqs)** **The formula for discounting is as follows: PV= ∑ Xi / (1+ r)i**

IMPORTANT EXAMPLE:

Where PV is present value
Xi is earnings from the investment in the year i
r is the rate of discount
Σ is the sum over i, of the discounted earnings.
Present value of machine that generates Rs. 1,000 for four years and then sold as scrap for Rs. 1,000 at the end of year 4?
Year 1+ Year 2 + Year 3 + Year 4
= 1000/ (1.1)1 + 1000/ (1.1)2 + 1000/ (1.1)3 + 2000/ (1.1)4
= 909+ 826+ 751 + 1366
= Rs. 3852
If machine costs less than this then Buy, otherwise don't Buy.
Net present value = PV – Purchase cost

(Lesson . 25)

Macroeconomics only began to be taught in colleges and universities in the **1940s** after the influence of a very influential **British** economist, **John Maynard Keynes**

MACROECONOMICS: Macroeconomics is a branch of economics that deals with the performance, structure, and behavior of a national economy as a whole.

AGGREGATE DEMAND (AD): The amount of total spending on domestic goods and services in an economy.

AGGREGATE SUPPLY (AS): The total amount of goods (including services) supplied by businesses within a country at a given price level.

- Below full employment Supply Curve (Flat) ho gi
- full employment Supply Curve (Steep) ho gi

CLASSICAL ECONOMICS:
Its major developers include Adam Smith, David Ricardo, Thomas Malthus and John Stuart Mill They were essentially micro-economists who believed the macro economy was an uninteresting aggregation of individual (or micro) markets, and any problem at the macro level was necessarily a symptom of some micro level problem.

OPTIMAL ROLE OF GOVERNMENT UNDER CLASSICAL ECONOMICS: classical economists developed theories of value, price, supply, demand,

and distribution. Nearly all rejected government interference with market exchanges, preferring a looser market strategy known as laissez-faire, or "let it be.

THE CONCEPT OF INVISIBLE HAND: Invisible hand was a concept introduced by Adam Smith in 1776 to describe the paradox of laissez-faire market economy.
how free markets can incentivize individuals, acting in their own self-interest, to produce what is societally necessary.

Demand Increase -------- Price Increase
Price Increase ------------ Demand Decrease

FULL EMPLOYMENT: when all available labor resources are being used in the most efficient way possible.

A. (Investment was low because the interest rate was too high in the loanable funds market. Policy recommendation: savings be increased to lower the interest rate and boost investment
B. Unemployment was high because of obstructions to the free market mechanism in the labour market which were preventing wages from falling to the market clearing level. Policy recommendation: these obstructions: benefit payments to unemployed, taxes on income and trade unions be eliminated) **FOR MCQS.**

KEYNES AND THE ORIGINS OF MODERN MACRO ECONOMICS:

Theory based on the ideas of the 20th-century British economist John Maynard Keynes. Keynesian economics promotes a mixed economy, in which both the state and the private sector are considered to play an important role. He argued that government policies could be used to promote demand at a macro level,

(Lesson . 26)

DIFFERENCES BETWEEN CLASSICAL AND KEYNESIAN ECONOMICS:
Three ways:

1. Labor market
2. Market for loanable funds
3. Aggregate demand and supply

LABOR MARKET: THE CLASSICAL VIEW: Excess labor supply lowers wages, prompting increased demand from firms, restoring market equilibrium

LABOR MARKET: THE KEYNESIAN VIEW: Excess labor supply and falling wages, as viewed by Keynes, lead firms to anticipate reduced consumer spending, discouraging new investments in production

MARKET FOR LOANABLE FUNDS: THE CLASSICAL VIEW:
Money market: Depositors supply loanable funds, while businesses borrow, creating the demand for loanable funds

MARKET FOR LOANABLE FUNDS: THE KEYNESIAN VIEW: Keynes: Boosting loanable funds raises savings, lowers consumer spending, and leads firms to cut new investments. The curve shifts downward, setting a new market-clearing interest rate at r2 due to reduced demand for investments caused by increased savings.

AGGREGATE DEMAND: Aggregate Demand (AD) represents the overall planned spending in the economy, combining consumption, investment, government spending, and net exports. It is inversely tied to the aggregate price level through wealth, interest rate, and international purchasing power effects. **AD curve slopes downward for both Keynes and classicals. Mcq**

WHY AD CURVE SLOPES DOWNWARD: (Three effects).

1. The interest rate effect of a price level increase on AD is negative
2. The wealth effect of a price level increase on AD is negative
3. The international purchasing power (or competitiveness)
4.

(Lesson . 27)

AGGREGATE DEMAND AND SUPPLY: THE CLASSICAL VIEW:
AS curve vertical; demand doesn't explain low activity. Policy: shift AS right (supply-side measures). Classicists: Always at full employment; new long-run equilibrium found automatically, no short-run discussion

According to classicals, economy is always at full employment level. Economy would automatically find the new equilibrium in the long run; they did not talk about short run. (For MCQs)

AGGREGATE DEMAND AND SUPPLY: THE KEYNESIAN VIEW: AS curve: Horizontal at less than full employment, then upward-sloping. Injection of demand in recession boosts output, employment, and income. Keynes: Output increases with price rise. Short run: Overtime possible, AS curve positively sloped; long run: AS curve becomes vertical.

Shifts in AD curve would have the impact on the output level. Output will increase as the AD curve shifts rightward. Keynes said that prices are fixed in the short run. (For MCQs)

KEYNES DEMAND MANAGEMENT POLICIES: Keynes exerted a phenomenal influence on economic thinking and policy-making in the 20th century and to date. In the 1950s and 60s,

Keynesian demand policies saw unemployment and inflation as opposites. Challenge: Unapplicable in stagflation—both rising prices and unemployment.

DIFFERENT SCHOOLS OF THOUGHTS

The Monetarist School: The Monetarist School, led by Milton Friedman
Inflation is a monetary phenomenon, says Milton Friedman's Monetarist School. To keep it low, align monetary growth with expected real output gr

The Real Business Cycles (RBC) School: The Real Business Cycles (RBC) School also gained currency in the 1970s. Business cycles proponents say that output changes are mainly due to technology shocks. They argue against using Keynesian policies to counter these effects.

The Rational Expectations School: The same period, 1970s
Rise of rational expectations school, led by Lucas, Barro, and Sargent. Agents use all available information, not just past, in decision-making. Their work shows predictable macro policies, like Keynesian demand management, have no impact on real output or unemployment.

Neo Classical Economics: Blend of monetarist and business cycle insights strengthens pre-Keynesian faith in free market power. Emphasizes micro-foundations of macroeconomics, labeled new or Neo-Classical Economics

The Neo Keynesian School:
Since the 1980s, the Neo-Keynesian School, led by economists like Joseph Stiglitz, emphasizes micro-level market failures due to issues like information gaps. They suggest government intervention to address problems like moral hazard and adverse selection, representing the current state of modern macroeconomics

(Lesson . 28)

Real data is nominal data adjusted for changes in prices. Most macroeconomic data is presented and analyzed in real terms so as to permit meaningful intertemporal and cross-country comparisons. (FOR MCQs)

Nominal Data	Real Data
Nominal data is the data which is expressed in monetary terms	Real data is the data after adjusting nominal data for inflation.

Gross Domestic Product (GDP): **(It is a Flow Measure)** Gross domestic product (GDP) is the value of the total final output produced inside a country, during a given year.

FLOW	STOCK
A flow figure refers to a certain period of time. Flow is referred to as the quantity that can be measured over a period of time.	A stock figure implies a particular point in time Stock refers to any quantity that is measured at a particular point in time

METHODS OF MEASURING GDP:

1. **The product or value added method** which sums the value added by all the productive entities in the economy;
2. **The expenditure method** which sums up the value of all the "final goods" transactions taking place in the economy;
3. **The factor income method** which sums up all the incomes earned by all the factors of production in the economy (rent for land, wages for labour, interest for capital, and equity returns for entrepreneurship).

Value added: An economic term to express the difference between the value of goods and the cost of materials or supplies that are used in producing them

Final Good	Intermediate Good
A good (or service) that is available for purchase by the ultimate or intended user with no plans for further physical transformation	A good (or service) that is used as an input or component in the production of another good.

(Lesson . 29)

Net Domestic Product (NDP): Net domestic product (NDP) measures the economic output of a country. It represents the net book value of all finished goods and services produced inside a country geographically during a given period.

NDP = GDP – Depreciation allowance

Depreciation: When capital is used over and over again to produce goods and services, it wears down from such use.

Gross National Product (GNP): The total value of goods and services produced by a country's citizens in a year, regardless of the location of production.

GNP = GDP + Net factor incomes from abroad

Net National Product (NNP): The total value of finished goods and services produced by a country's citizens overseas and domestically, minus depreciation.

NNP = GNP – Depreciation allowance

National income (NI): The total market value of production in a country's economy during a year. (**NNP is often referred to as national income). For MCQs**

Real GDP	Nominal GDP
Real GDP measures the economic production factoring in any prices changes in the market (deflation or inflation).	Nominal GDP measures the economic production at current market prices, The total value of all goods and services produced in a given time period less the value of those made during the production process.

PRICE DEFLATOR: The process of converting nominal GDP into real GDP is known as deflation

GDP Deflator = Nominal GDP / Real GDP

Per Capita GDP: Per capita GDP is simply the total GDP of the economy divided by the no. of people in the economy
THE PURCHASING PARITY (PPP): the rates of currency conversion that try to equalise the purchasing power of different currencies, by eliminating the differences in price levels between countries.

Personal Income	Disposable Income
Personal income includes payments to individuals (income from wages and salaries, and other income), plus transfer payments from government, less employee social insurance contributions.	Disposable personal income measures the after-tax income of persons and nonprofit corporations.

DRAWBACK OF GDP BASED MEASURES:

1. GDP does not capture welfare or human well-being.
2. GDP may not be a strong basis to predict economic growth in times of high uncertainty
3. As international accounting standards are slow to change and require international consensus,
4. GDP is slow to reflect changes in the world.

(Lesson . 30)

THE KEY VARIABLES OF THE MACROECONOMIC MODEL:

Factor services supplied by households **(MCQs)**

- Y = Income
- C = Consumption
- S = Savings
- I = Investment
- T = Taxes
- G = Government Expenditures
- M = Imports
- X = Exports

CIRCULAR FLOW: The circular flow means the unending flow of production of goods and services, income, and expenditure in an economy.

Injection:	**Leakage/Withdrawals:**
Injections are the introduction of income into the flow, such as additions to investment, government expenditure and exports.	Leakages are the withdrawal of income from the flow, such as savings, taxation and imports.

Injection-leakage model used to identify equilibrium aggregate output in Keynesian economics. (FOR MCQs)

Injection-leakage model: A model used in Keynesian economics based on the equality of non-consumption expenditures (or injections) and non-consumption uses of income (leakages).

MACROECONOMIC EQUILIBRIUM: CLASSICAL VIEW: When all sections are equal Saving equals to Investment or Tax equals to Expenditure or Import equals to Exports Equilibrium is achieved when all condition are equal.

MACROECONOMIC EQUILIBRIUM: KEYNESIAN VIEW: When total injections equal total leakages (or total withdrawals), or aggregate supply equals aggregate demand. Equilibrium is achieved.

Withdrawal = Injection

S + T + M = I + G + X **For MCQs**

Y = AD = AS

Income = Expenditure = Output **For MCQs**

Aggregate supply is a function of available inputs, technology and the price level. (MCQs)

Consumption is the most stable and important component of aggregate demand, accounting for about two-thirds to three-fourths of GDP in most countries. (MCQs)

Important Question:

Disposable Income (Yd)	Consumption (C`)	Saving (S = Yd – C`)
500	**500**	**0**
550	**540**	**10**
600	**580**	**20**
650	**620**	**30**
700	**660**	**40**
750	**700**	**50**
800	**740**	**60**

Marginal Propensity to Consume (MPC) and Marginal Propensity to Save (MPS): Marginal propensity to consume (MPC) is the extra amount that people consume when they receive an extra dollar of disposable income. MPC's numerical value is usually between 0.5 and 1

Marginal propensity to save (MPS) is the fraction of the additional dollar of disposable income that is saved. **(Formula): MPC = 1 – MPS**

Average propensity to consume (APC) or Average propensity to save (APS): The ratio of total consumption to total disposable income **(APS)** is the ratio of total saving to total disposable income. **(Formula): APC = 1 – APS.**

THE SAVING FUNCTION: The saving function yields the amount of saving that households of a nation will undertake at each level of income. Formula is S = c + d(Yd). d is MPS, positive, and usually less than 0.5.
The relationship between saving and the interest rate is also important. The relationship is positive,(for Household) **(For MCQs)**

CALCULATION OF APC, APS & MPC, MPS

- **Average propensity to consume:**
  APC = C / Yd
- **Average propensity to save:**
  APS = S / Yd
  Or
  APS = 1 – APC
- **When** **Yd** = \$500 billion **APC** = 1 **APS** = 0
- **When** **Yd** > \$500 billion **APC** < 1 **APS** > 0
- **Marginal propensity to consume:**
  MPC = ΔC / ΔYd
- **Marginal propensity to save:**
  MPS = ΔS / ΔYd
  Or
  MPS = 1 – MPC

APC = C/Yd	APS	Yd	C	MPC(ΔC/ΔYd)
500/500=1.0	0	500	500	
540/500=0.98	0.02	550	540	40/50=0.8
580/600=0.97	0.03	600	580	40/50=0.8
620/650=0.95	0.05	650	620	40/50=0.8
660/700=0.94	0.06	700	660	40/50=0.8
700/750=0.93	0.07	750	700	40/50=0.8
740/800=0.92	0.08	800	740	40/50=0.8

(Lesson . 31)

INVESTMENT AND INVESTMENT DEMAND CURVE: The investment demand curve shows the relationship between the level of investment and the cost of borrowing for the firm (i.e. the interest rate), plotted in i-I space. The relationship between the interest rate on such borrowing and investment demand is obviously negative, **(MCQs)**

The interest rate goes up, investment demand decreases. **(MCQs)**

TYPES OF INVESTMENT:

1. Residential construction
2. Non-residential construction
3. The demand for producers
4. Changes in business inventories

Imports	Exports
Imports are goods and services that are produced in another country and consumed in the home country.	Exports are goods and services that are produced in the home country and consumed in another country.

(Quotas) Limit karta ha imports ko

Trade balance: is the excess of exports over imports. A negative trade balance is called a trade deficit **(MCQs)**

Fiscal Policy: is the use of government spending and taxation to influence the economy.

T-G is referred to as the fiscal balance. If G>T, there is a fiscal or budget deficit; if G<T, there is a fiscal or budget surplus. If G=T, there is a balanced budget. **(MCQs)**

An inflationary gap refers to a situation where there is pressure on prices to rise. **(MCQs)**

Injection curve (Upward Sloping)
Leakage curve (Horizontal Sloping) (MCQs)
The term **[1/(1-b)]** is called the Keynesian multiplier

(Lesson . 33)

Four major problems here:

1. Unemployment
2. Inflation
3. Balance of payments problem
4. The lack of growth

DEFINITION OF UNEMPLOYMENT: when someone is willing and able to work but does not have a paid job. The unemployment rate is the percentage of people in the labour force who are unemployed.

A rate of 3-4% is usually considered low, 10-15% considered high, and over 20% considered extremely high

Types of Unemployment:

1. **Frictional unemployment:** When workers are searching for new jobs or are transitioning from one job to another.
2. **Structural unemployment:** Because workers lack the requisite job skills or live too far from regions where jobs are available and cannot move closer
3. **Cyclical unemployment:** Unemployment which is caused by the changes in business cycles, that is economic upturn or downturn.

4. **Technological unemployment:** Is caused by the replacement of workers by machines or other advanced technology.
5. **Classical or real-wage unemployment:** When real wages, or the cost of employing a worker, are too high

The Concept of Labor Force (LF): The labor force comprises individuals who are either employed or actively seeking employment. It does not include those who are not working and not actively seeking a job.

People entering the unemployed pool include: People entering the unemployed pool are individuals becoming unemployed, such as those who lost their jobs.

People leaving the unemployed pool include: people leaving the unemployed pool are those transitioning from unemployment to employment, indicating a reduction in the unemployed population.

THEORIES ABOUT THE CAUSES OF UNEMPLOYMENT:

THREE VIEWS ABOUT UNEMPLOYMNET:

- Classical School
- Keynesian School
- Monetarist school

CLASSICAL VIEWS: Unemployment depends on the level of real wages. It occurs when real wages are fixed over the equilibrium level because of rigidities provoked by minimum-wage policies, union bargaining or effective salaries.

KEYNESIAN VIEWS: Unemployment in deficient aggregate demand. According to him Keynes argued that inadequate overall demand could lead to prolonged periods of high unemployment.

Seasonal unemployment: when people who work in seasonal jobs become unemployed when demand for labor decreases.

Monetarists also suggested supply side measures which generally reduced the incentive to work. One example is lowering income tax rates and thus increasing the incentive for people to work the extra hour and earn the extra dollar.